

DATA ENGINEER

WEEK1: DELIVERABLE DOCUMENT

Smart Water Quality surveillance & Loss Detection System

★ Task - 01

1) Ingestion Schema Definition:

Field Name	Data Type	Unit	Nullable?	Example Value	Source	Description & Validation Rules
sensor_id	String	N/A	No	SNS-LOC01-001	IoT ESP32	Unique sensor identifier. Format: SNS-{location}-{seq}. Must match IoT registry.
timestamp	DateTime	ISO 8601	No	2025-05-10T09:15:00Z	ESP32 RTC / NTP	UTC timestamp of reading. Must be ISO 8601 format. Reject if null or malformed.
location_id	String	N/A	No	ZONE-A	IoT config	Physical zone/pipe identifier. Must match location master list agreed with IoT.
pH	Float	pH unit	No	7.2	pH Sensor	Water acidity. Valid range: 0.0–14.0. Alert threshold: <6.5 or >8.5.
turbidity	Float	NTU	No	3.5	Turbidity Sensor	Water cloudiness. Valid range: 0–1000 NTU. Alert threshold: >4 NTU (WHO).
TDS	Float	mg/L (PPM)	No	210.0	TDS Sensor	Total Dissolved Solids. Valid range: 0–2000 mg/L. Alert threshold: >500 mg/L.
DO	Float	mg/L	No	8.1	DO Sensor	Dissolved Oxygen. Valid range: 0–20 mg/L. Alert threshold: <6 mg/L.
flow_rate	Float	L/min	No	15.3	Flowmeter	Water flow rate. Valid range: 0–500 L/min. Used for leakage detection logic.

pressure	Float	kPa	No	101.5	Pressure Sensor	Pipe pressure. Valid range: 0–600 kPa. Sudden drop may indicate leakage.
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2) Schema Validation Rules (for ingestion script):

Rule Type	Field(s)	Condition	Action on Fail
Null check	All fields	No null values permitted	Reject record, log error
Type check	All numeric fields	Must parse as float/int without error	Reject record, log error
Range check	pH	$0.0 \leq \text{pH} \leq 14.0$	Flag as anomaly, store with alert flag
Range check	turbidity	$0 \leq \text{turbidity} \leq 1000 \text{ NTU}$	Flag as anomaly, store with alert flag
Range check	TDS	$0 \leq \text{TDS} \leq 2000 \text{ mg/L}$	Flag as anomaly, store with alert flag
Range check	DO	$0 \leq \text{DO} \leq 20 \text{ mg/L}$	Flag as anomaly, store with alert flag
Format check	timestamp	Must parse as valid ISO 8601 UTC datetime	Reject record, log error
Uniqueness	sensor_id + timestamp	Combination must be unique per record	Reject duplicate, log warning

★Task - 02: Time-Series Storage Setup

1) Storage Options: Technical Comparison:

Criteria	InfluxDB	Parquet Flat-File Store
Best for	Real-time time-series data at high frequency	Batch analysis, historical queries, ML training
Query language	Flux / InfluxQL — purpose-built for time-series	SQL via DuckDB / Pandas — familiar to most
Write speed	Very fast — optimised for high-frequency inserts	Moderate — writes batched to Parquet files
Read/query speed	Fast for recent/time-range queries	Very fast for large historical range scans
Setup complexity	Medium — install InfluxDB server, configure buckets	Low — just write Parquet files to disk
Retention policies	Built-in — auto-delete old data after X days	Manual — manage files yourself
Dashboard support	Native Grafana integration out of the box	Needs additional tool (e.g. Plotly, Streamlit)
Partitioning	By measurement name + tags (location, sensor type)	By folder structure: /location/sensor_type/date/
Recommended for us	YES — real-time monitoring + alert system alignment	GOOD for Data Science team's ML pipeline (supplementary)

2) InfluxDB Configuration Plan:

▪ Partitioning Structure:

- Measurement name = sensor type (e.g. 'water_quality', 'flow_data')
- Tag: location_id (e.g. 'ZONE-A', 'ZONE-B') — enables filtering by pipe/zone
- Tag: sensor_id (enables per-device queries)
- Fields: pH, turbidity, TDS, DO, flow_rate, pressure (numeric values)
- Timestamp: auto-indexed by InfluxDB using the timestamp field from schema

▪ Bucket Setup:

Bucket Name	Purpose	Retention Policy
water_realtime	Live sensor readings from ESP32	30 days (then archive to Parquet)
water_archive	Historical data for Data Science team	Infinite (Parquet files on disk)